

Heritage Building Information Modelling in Group Portrait
Depiction of Regional Historical Wooden Buildings: from
Beiyuyuan Five-Mountain Temple to Fenyang

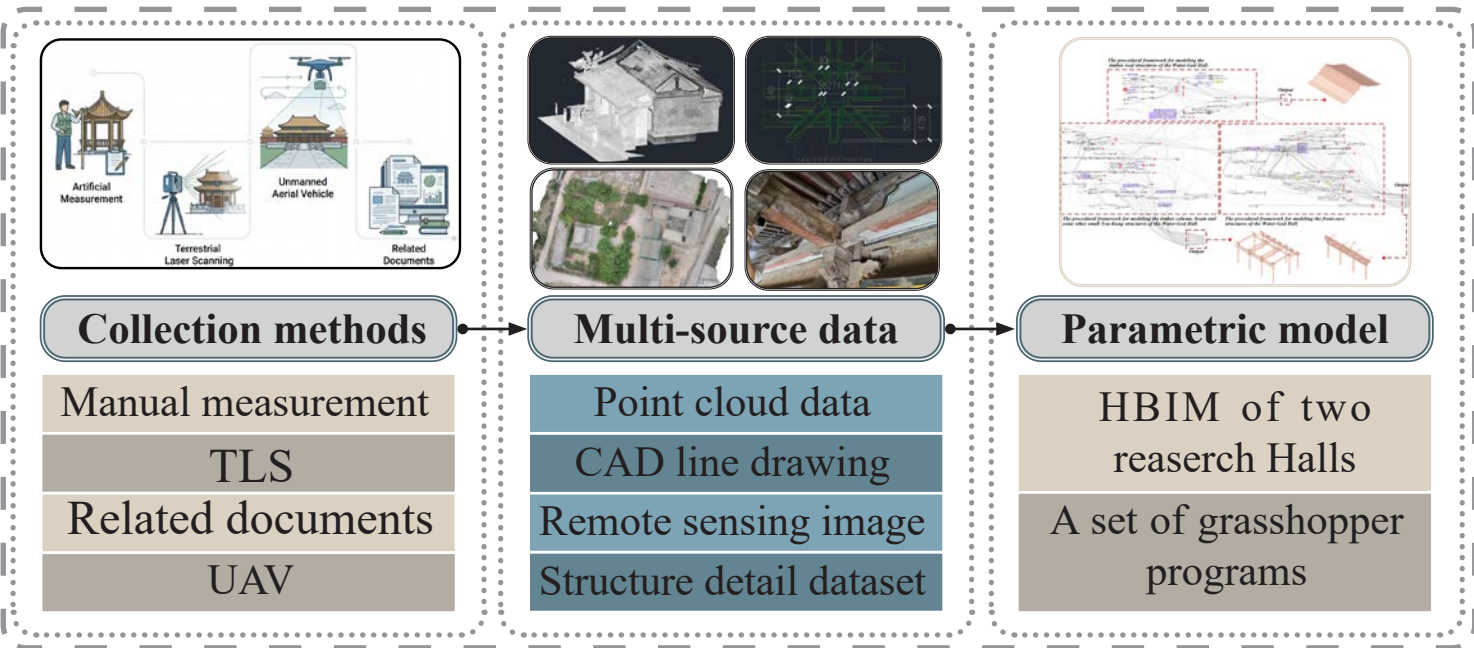
Abstract

This study explores a **Historical Building Information Modeling (HBIM)** workflow to establish a parametric component library for Yuan Dynasty timber buildings in the **Fenyang area**. Starting with the **Water-God Hall** at Beiyuyuan Five-Mountain Temple as a case study, **terrestrial laser scanning (TLS)** and **unmanned aerial vehicle (UAV)** data were integrated with on-site measurements to create accurate point cloud models. Using Rhino and Grasshopper for parametric modeling, a reusable component library was developed. The model was transferred to Autodesk Revit via Rhino.Inside to incorporate non-geometric information, completing the HBIM. This model was then efficiently adapted to other structurally similar buildings, such as the **Five-Mountain Hall**, by adjusting parameters. The research demonstrates an effective pipeline for digital heritage conservation, enabling comparative architectural analysis and providing a scalable, shareable resource for studying regional structural characteristics.

Keywords

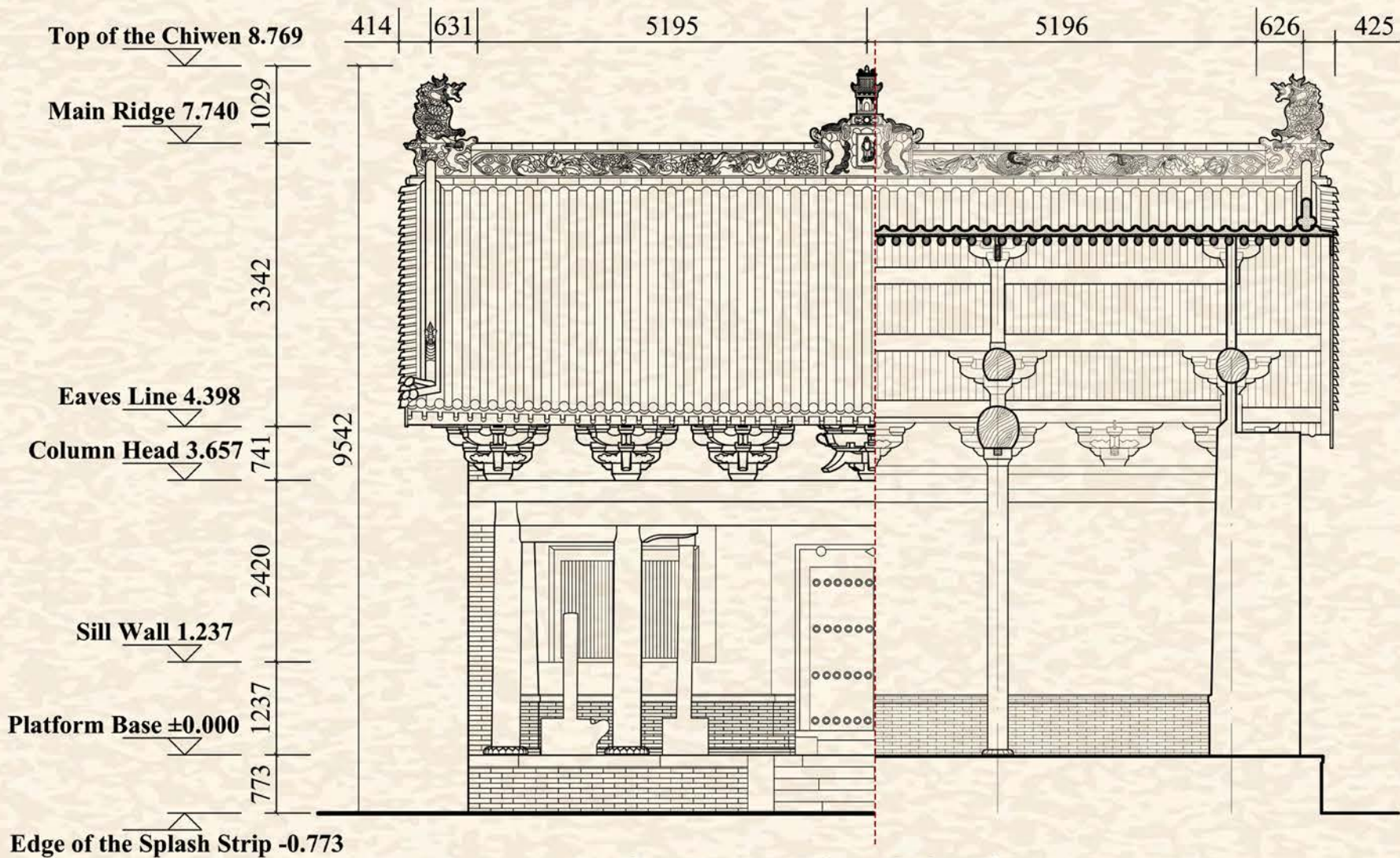
Chinese Architecture	HBIM	Digital Conservation
Tou-Kung	Parametric Design	Yuan Architecture

Technology Roadmap

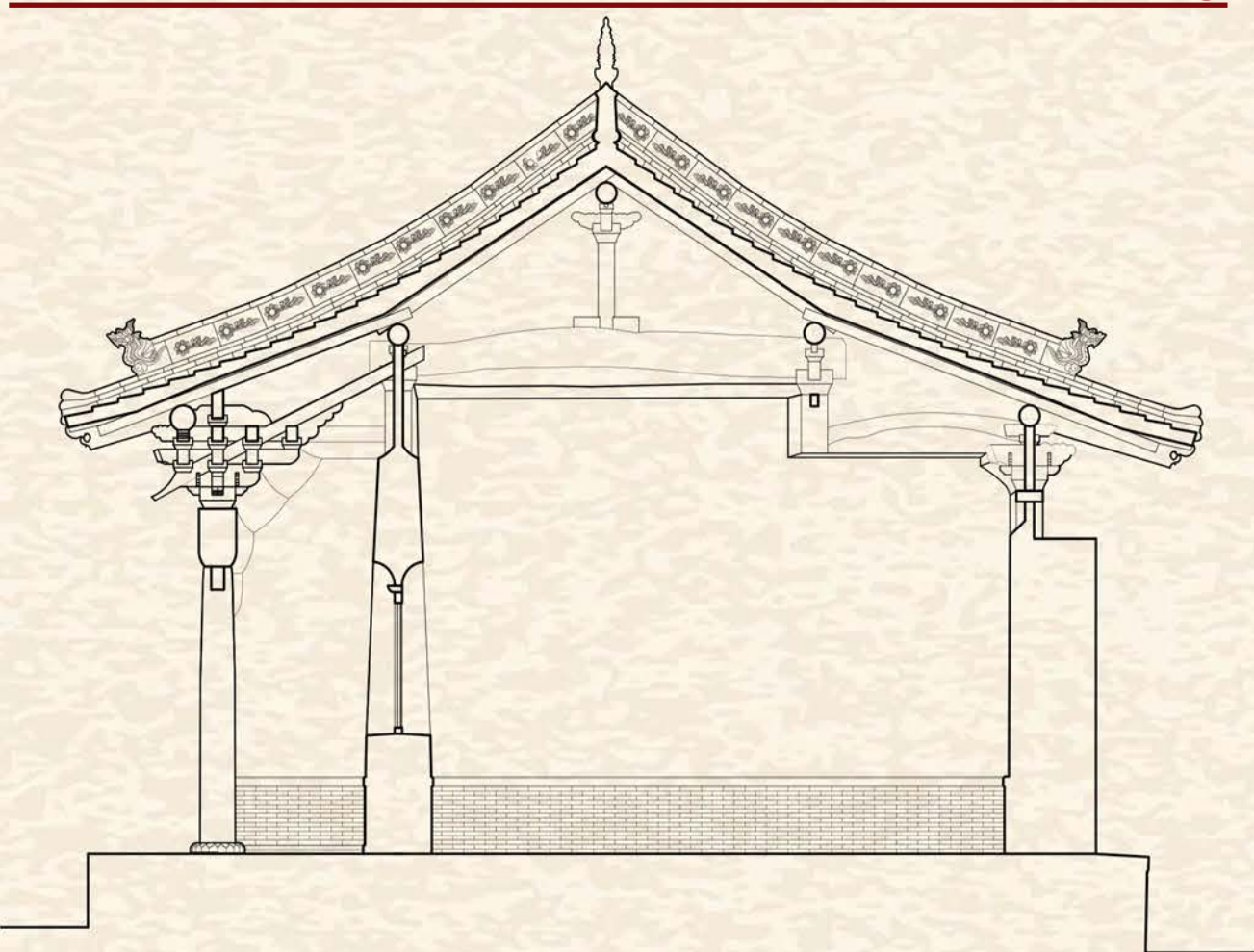


Site plan (Remote sensing image mosaic by UAV)



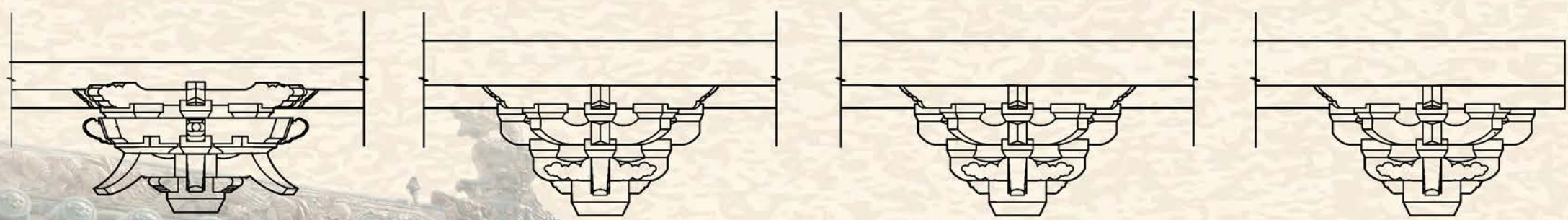


South elevation & Section



Section of the secondary bay

Bracketing set elevation

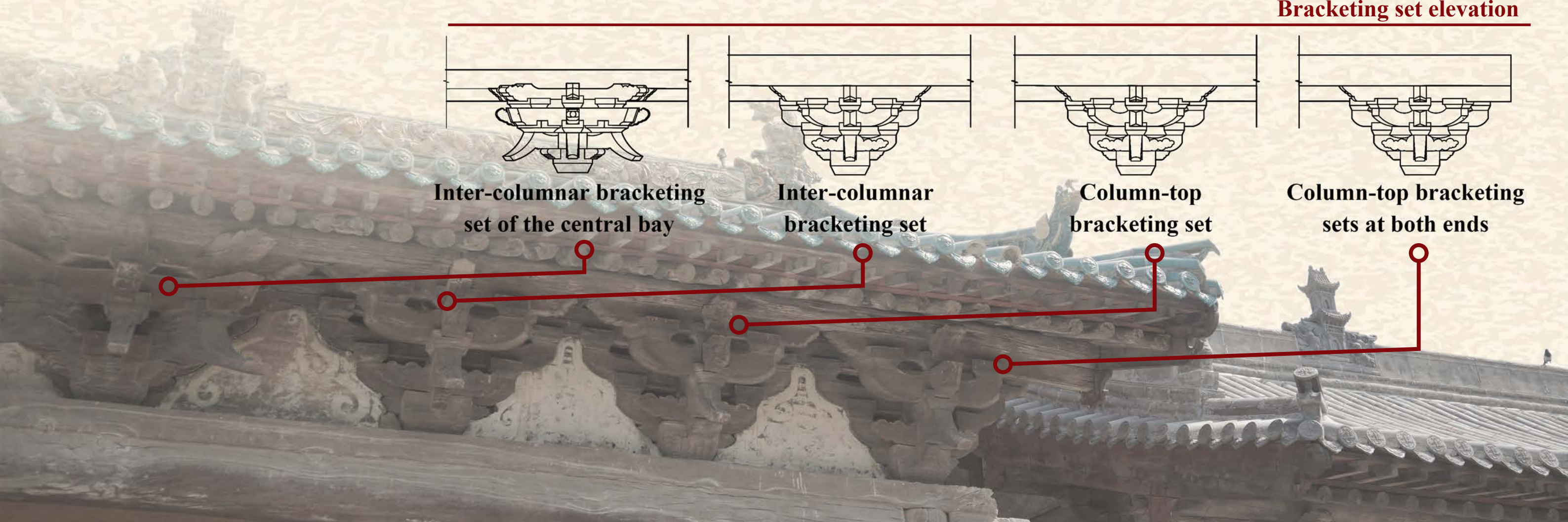


Inter-columnar bracketing set of the central bay

Inter-columnar bracketing set

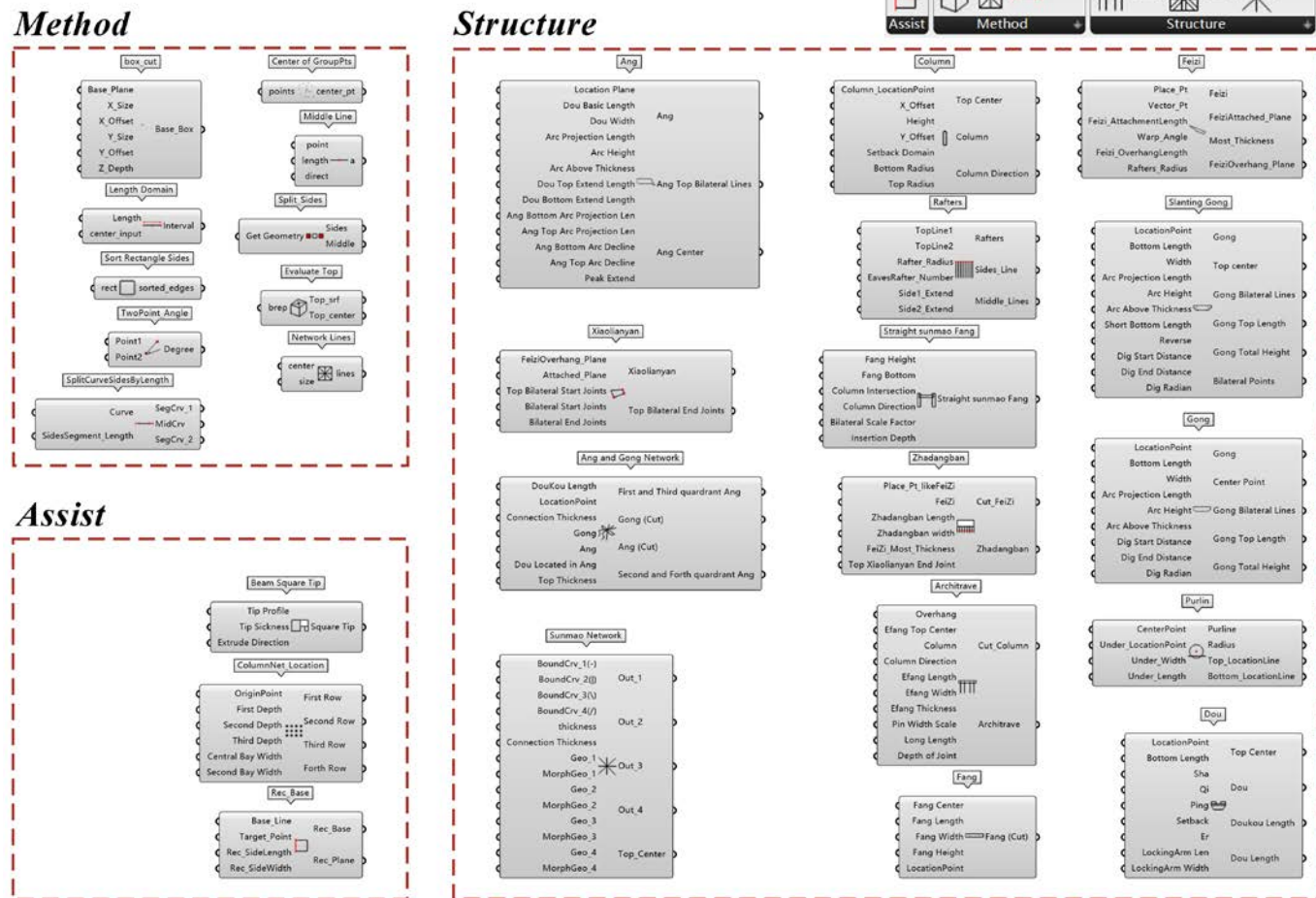
Column-top bracketing set

Column-top bracketing sets at both ends

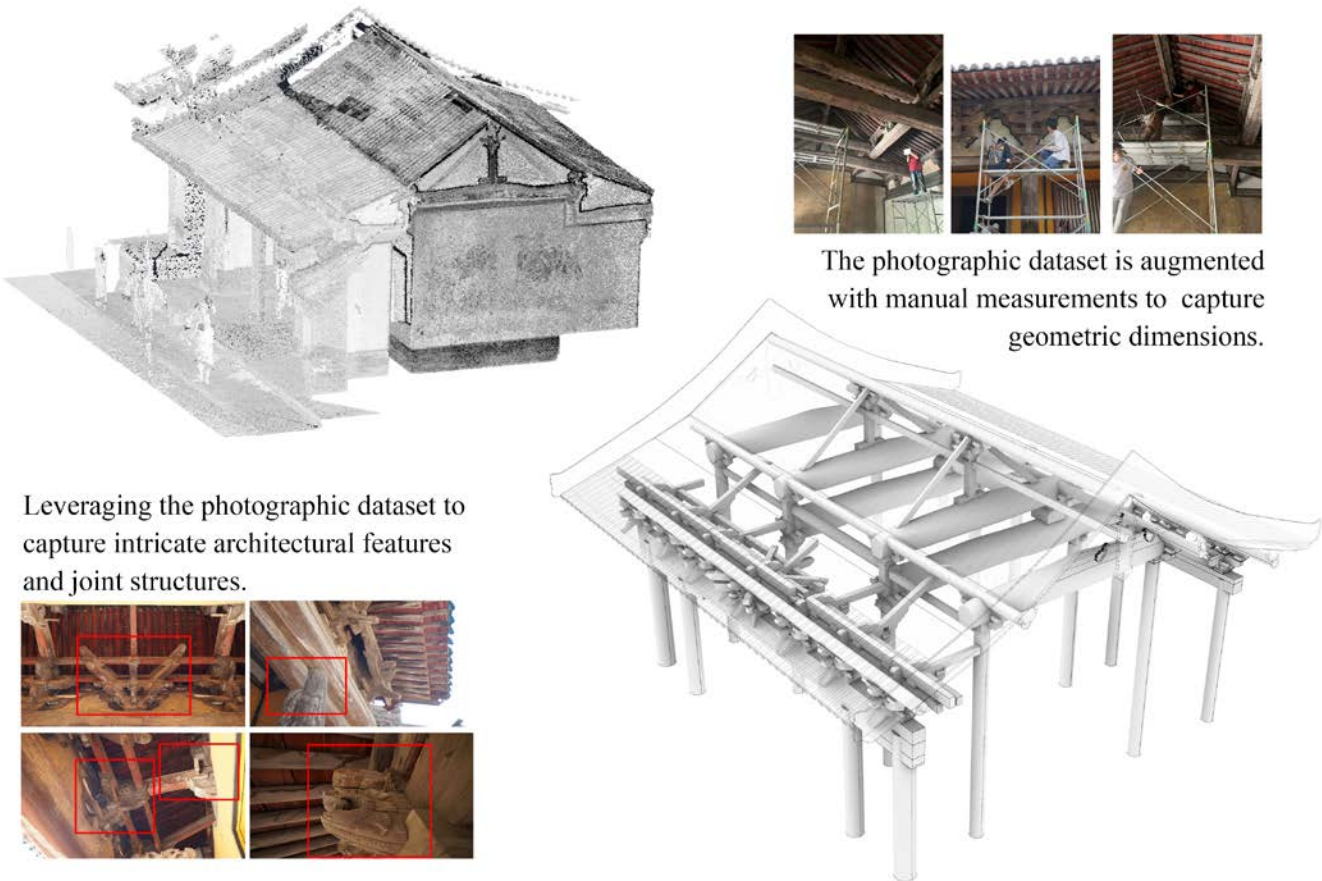


Contributing relevant modeling program components

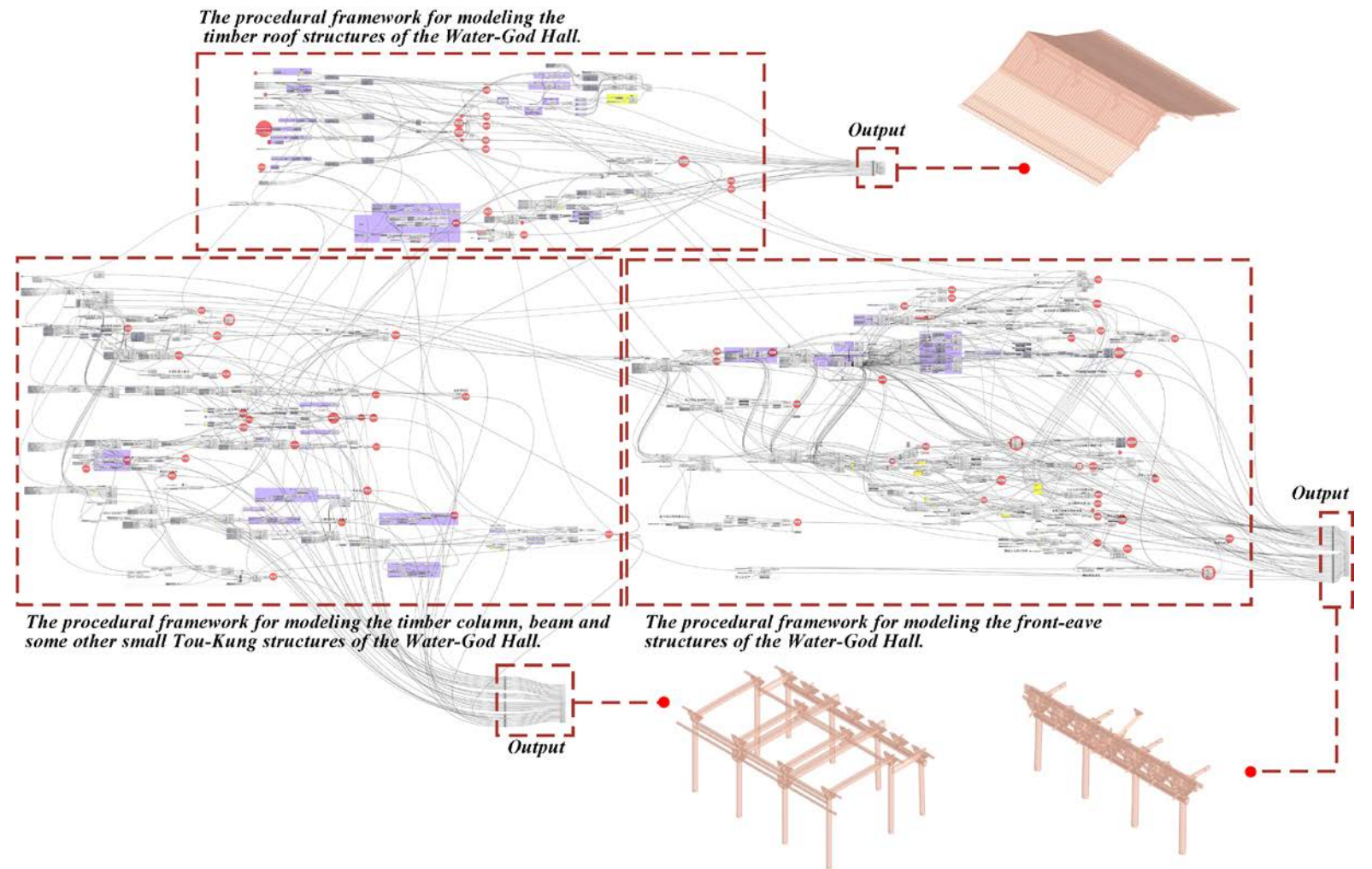
HBIM Program Components for Beiyuyuan



Point cloud-based 3D reconstruction and reverse engineering



Parametric program contribution



The comparison of timber structures of two halls

Extracting differences and commonalities within the same region to uncover the typological patterns of local ancient architecture.

